# Presentation-aware E-values, cross-allele pseudosequence diffusion, and proteome-scale peptide matching for peptide–MHC analysis

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*mhcmatch technical appendix*

**Abstract**

This appendix develops the mathematical and statistical theory of `mhcmatch`, the applied peptide–MHC layer built on the `seqtree` fuzzy-search substrate. We take as given the control-calibrated E-value of the `seqtree` appendix (`evalue.tex`) and specialize it to MHC *presentation*, whose null is allele-conditional and anchor-constrained. We restate the anchor / TCR-facing decomposition and the per-allele presentation-aware E-value (the *forward* restriction problem), then the *reverse* problem—peptide to presenting allele—as a neighbour-vote posterior with a binomial-tail confidence E-value, together with its non-binder filter and class-II promiscuity treatment. The central contribution is a *cross-allele diffusion* model: each allele is represented by its 34-residue groove pseudosequence, and per-anchor preference distributions are smoothed by kernel-weighted empirical-Bayes shrinkage toward groove-similar alleles, with the per-anchor relevance of each groove position *learned* from data (a mutual-information feature importance that separates, e.g., the MHC-I B-pocket governing P2 from the F-pocket governing PΩ). This rescues rare alleles with few peptides—lifting the `seqtree` limitation that distinct alleles are distinct, never-pooled nulls—and we give its hierarchical-Bayes interpretation, effective sample size, limiting cases, and bias–variance trade-off. We then treat proteome-scale scanning (sliding-window presentation with multiple-testing control; near-exact neoantigen source identification), motif logos with length distributions, the composition rules for the planned stability / affinity / cleavage / expression / immunogenicity predictors, and the benchmark protocol against NetMHCpan, NetMHCIIpan, MixMHCpred and MixMHC2pred.

## 1. Scope and relation to the substrate

`seqtree` provides a payload-agnostic fuzzy-search core and a control-calibrated E-value: for a query $q$ and a scope $\theta$, with an empirical background $P_0$ estimated from a matched control of size $M$ and

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a target set of size $N$, the expected number of chance neighbours is $\widehat{E}(q,\theta) = (N/M)\, n_C(q,\theta)$, with $p_{\text{any}} = 1 - e^{-\widehat{E}}$ and $p_{\text{enrich}} = \mathbb{P}(\text{Poisson}(\widehat{E}) \geq n_D)$ (`seqtree` appendix, Def. of the E-value; Poisson/Chen–Stein bound; Karlin–Altschul [9, 1] as the i.i.d. special case). We do not re-derive that theory; we *specialize* it.

For peptide–MHC the relevant background is not V(D)J generation but *presentation*, which is allele-conditional and constrains the anchor positions. `mhcmatch` owns four developments on top of the substrate: (i) the productionized forward/reverse presentation E-value (§ 2–3); (ii) the cross-allele *pseudosequence diffusion* model that pools statistics across groove-similar alleles (§ 4, the headline); (iii) proteome-scale scanning and near-exact source identification (§ 5); and (iv) motif logos (§ 6). Planned predictors and the comparison protocol are § 7–8; limitations are § 9.

## 2. Anchor decomposition and the per-allele presentation E-value

A presented peptide of length $L$ factorizes $\sigma = (\sigma_A, \sigma_T)$ into MHC-facing *anchors* $A(a, L)$ and TCR-facing positions $T$. For class I the buried anchors are $\{\text{P2}, \text{P}\Omega\}$ (the B- and F-pockets); for the class II 9-mer core they are $\{\text{P1}, \text{P4}, \text{P6}, \text{P9}\}$. *Presentation* fixes $\sigma_A$; *recognition* reads $\sigma_T$. Dolton et al. [6] exhibit one HLA-A02:01 TCR cross-reacting with three epitopes through a shared central motif with the anchors irrelevant, which motivates two complementary maskings, each realized by a per-position weight vector $w$ over a `seqtree PositionalMatrix`:

$$\text{(recognition)} \quad w_i = \mathbb{1}[i \in T]; \qquad \text{(presentation)} \quad w_i = \mathbb{1}[i \in A(a, L)]. \tag{1}$$

In `mhcmatch` these are exposed by `Store.decompose`, which returns both readouts with the masked positions written as the spare symbol `X`.

**Forward problem..** For a candidate restriction allele $a$ with an anchor-masked presented control $C_a$ of size $M_a$ and target peptidome of size $N_a$, the presentation-aware E-value is the substrate E-value on allele-restricted, anchor-conditioned indices,

$$\widehat{E}(q, \theta; a) = \frac{N_a}{M_a}\, n_{C_a}(\sigma_T, \theta), \qquad p_{\text{enrich}} = \mathbb{P}(\,\text{Poisson}(\widehat{E}) \geq n^T_{D_a}), \tag{2}$$

i.e. significance is computed on the TCR-facing readout only, so shared anchors (presentation, not recognition) do not inflate within-allele hits. The analytic null for rare $a$ is taken up in § 4.

## 3. Reverse problem: peptide to presenting allele

The forward E-value answers "is $q$ a binder of a *given* $a$?". Restriction prediction asks the reverse: which alleles present $q$? We flip the weights in (1) to the *presentation* mode and index reference peptides by their anchor signature (`layout.presentation_features`; class I N/C-pocket residues P1 P2 P3, P(Ω—1), PΩ, class II core P1 P4 P6 P9). For a query we widen the scope on its signature until it has $n \in [10, 100]$ non-exact neighbours, and tally their alleles, $k_a$ votes out of $n$.

Definition 1 (Ranking and confidence). The *ranking* score is the neighbour vote fraction $\hat{p}(a \mid q) = k_a/n$, a posterior over the panel. The *confidence* score is the per-allele enrichment over the panel background frequency $f_a$,

$$E_a = -\log_{10} \mathbb{P}(\,\text{Binomial}(n, f_a) \geq k_a), \tag{3}$$

the upper binomial tail; a peptide binds $a$ iff $E_a \geq -\log_{10}\alpha$ and $k_a > 0$.

Remark 1 (Why vote fraction, not enrichment, ranks). On a skewed panel the dominant true allele has a large expected count $nf_a$, so the enrichment (3) *penalises* it; the vote fraction does not. Hence $\hat{p}$ ranks and $E_a$ gates. This separation, validated on `pmhc_data` (per-(peptide, allele) ROC-AUC 0.90–0.99, `seqtree/bench/bench_mhc_guess.py`), is what `Store.restriction` implements.

**Class-II register trick..** The class-II core register inside a longer peptide is unknown; full deconvolution clusters it [3, 2]. We use a one-pass proxy: pick the single 9-mer window whose P1 is a large hydrophobic and whose P4/P6/P9 avoid Pro/Gly, and take that core's {P1,P4,P6,P9}. Committing to one register makes an allele's peptides share a consistent signature and is what lifts class-II ROC-AUC into the 0.9 range.

**Non-binder filter and promiscuity..** A peptide that binds *no* panel allele has small $\max_a E_a$ (real-vs-random rejection); one that does not bind a *specific* $a$ has $E_a < -\log_{10}\alpha$. The global statistic $E_{\text{glob}} = \sum_a 10^{-E_a}$ is the expected number of panel alleles presenting $q$ by chance. Class II is genuinely multi-label (an open groove with loosely specified pockets presents one peptide on many alleles), so the panel positive set is *every* observed restricting allele and the non-binder test is "binds none of the panel".

## 4. MHC pseudosequence model and cross-allele diffusion

`seqtree` deliberately treats distinct alleles as distinct, never-pooled nulls. This is correct but wasteful: rare alleles carry few presented peptides, so both (2) and (3) are noisy exactly where data are thin—the equity problem of Glynn et al. [8]. `mhcmatch` lifts the limitation by making the allele a *sequence* the same engine can compare, and diffusing statistics along groove similarity.

**Pseudosequence..** Each allele $a$ is represented by its NetMHCpan-style 34-residue groove pseudosequence $s_a \in \Sigma^{34}$ [14]: the polymorphic, peptide-contacting positions (class I $\alpha_1/\alpha_2$; class II $\alpha_1 + \beta_1$). Per-allele binding can be predicted from $s_a$ alone [8, 4], so groove distance is a meaningful allele distance.

### 4.1. Anchor-factored similarity kernel

The presentation motif factorizes by pocket: anchor $j$'s residue preference is governed by the groove positions lining pocket $j$, and different pockets are spatially separated (the class-I B-pocket sets P2, the F-pocket sets PΩ). We therefore use a *per-anchor* kernel rather than one global allele distance. For anchor $j$ let $w_j \in \mathbb{R}^{34}_{\geq 0}$ weight the groove positions, and define

$$d_j(a,b) = \sum_{p=1}^{34} w_{j,p}\, \mathbb{1}[s_{a,p} \neq s_{b,p}], \qquad K_j(a,b) = \exp(-d_j(a,b)/h_j), \tag{4}$$

a weighted-Hamming kernel with bandwidth $h_j > 0$ (ambiguous positions `X` are skipped). Setting $w_j \equiv 1$ recovers a single un-factored kernel.

Definition 2 (Learned per-anchor relevance). Let $A_j$ be the allele's modal residue at peptide anchor $j$ (from its presented set) and $S_p$ the groove residue at pseudosequence position $p$, both viewed as categorical variables over the allele panel. The relevance of position $p$ to anchor $j$ is the mutual information

$$w_{j,p} \;\propto\; \mathrm{I}(S_p; A_j) = \sum_{x,y} \hat{\mathbb{P}}(S_p{=}x, A_j{=}y)\, \log_2 \frac{\hat{\mathbb{P}}(S_p{=}x, A_j{=}y)}{\hat{\mathbb{P}}(S_p{=}x)\,\hat{\mathbb{P}}(A_j{=}y)}, \tag{5}$$

normalized to mean one over $p$. Equivalently $w_j$ may be fit by maximizing the held-out predictive log-likelihood of the shrinkage model below under an $\ell_1$/entropy penalty, which selects each anchor's pocket positions. (5) is the "feature importance" that says which groove residues drive P2 versus PΩ.

### 4.2. Kernel-shrinkage pooling

Write $\theta_{a,j}$ for allele $a$'s empirical residue distribution at anchor $j$ over its $n_a$ peptides. We shrink it toward groove-similar alleles:

$$\hat\theta_{a,j} = \frac{n_a\,\theta_{a,j} + \sum_{b\neq a} K_j(a,b)\,n_b\,\theta_{b,j}}{n_a + \sum_{b\neq a} K_j(a,b)\,n_b}, \qquad \mathrm{ESS}_{a,j} = n_a + \sum_{b\neq a} K_j(a,b)\,n_b. \tag{6}$$

Proposition 1 (Hierarchical-Bayes reading and limits). *(6) is the posterior mean of a Dirichlet–multinomial in which the neighbour-weighted counts form the prior pseudocounts: $\hat\theta_{a,j} = \mathbb{E}[\theta \mid \text{counts}]$ with concentration $\mathrm{ESS}_{a,j}$. As $h_j \to 0$, $K_j(a,b) \to 0$ for $b \neq a$ and $\hat\theta_{a,j} \to \theta_{a,j}$ (no pooling); as $h_j \to \infty$, $K_j(a,b) \to 1$ and $\hat\theta_{a,j} \to (\sum_b n_b\theta_{b,j})/\sum_b n_b$ (the global pool).*

(Both limits are implemented and unit-tested in `Pseudoseq.shrink`.)

Proposition 2 (Bias–variance and bandwidth). *Treat each $\theta_{b,j}$ as an unbiased estimate of allele $b$'s truth $\theta^\star_{b,j}$ with variance $\propto 1/n_b$. The pooled estimator has variance $O(1/\mathrm{ESS}_{a,j})$ and bias $\|\sum_b \kappa_b(\theta^\star_{b,j} - \theta^\star_{a,j})\|$, where $\kappa_b = K_j(a,b)n_b/\mathrm{ESS}_{a,j}$ is the borrowed mass. The mean-squared error is minimized by borrowing more (smaller effective $h_j$ penalty) when $n_a$ is small and groove neighbours are close, and less when $n_a$ is large; calibrating $h_j$ by cross-validated predictive likelihood realizes this trade-off automatically.*

**Bounded-concentration variant (the practical default).**. The counts-weighted form (6) lets one deep neighbour ($K_j n_b \gg n_a$) overwrite a rare allele's own peptides. We therefore default to a fixed prior strength $\tau$: shrink toward the kernel-weighted neighbour mean $m_{a,j} = (\sum_b K_j(a,b)n_b\theta_{b,j})/\sum_b K_j(a,b)n_b$,

$$\hat\theta_{a,j} = \frac{n_a\,\theta_{a,j} + \tau\,m_{a,j}}{n_a + \tau}, \tag{7}$$

so borrowing adds at most $\tau$ pseudocounts and vanishes automatically as $n_a$ grows (Prop. 2). Empirically (`bench/bench_diffusion.py`; MHC-I human, allele-split held-out rank AUC, $\tau = 10$) this lifts the AUC of *rare* alleles ($\leq 30$ peptides) from 0.87 to 0.92 on the high-confidence ($\geq 2$-publication) set while leaving *frequent* alleles unchanged ($\pm 0.00$); on the noisier full set the rare-allele gain is smaller but stays non-negative and frequent alleles are never harmed—the data-rescue is safe.

**Pooled null and restored E-value..** For a rare allele the analytic presented null is built from the shrunk anchor marginals: assuming approximate position independence in the anchor block (the maximum-entropy presented distribution of [15] conditioned on the motif), $\widehat{P_0}(\sigma_A \mid a) \approx \prod_j \hat{\theta}_{a,j}(\sigma_{A_j})$, which feeds a usable $\widehat{E}$ in (2)–(3) where the raw per-allele control was too small. The estimator is a smoothed null: its bias is controlled by the borrowed mass on dissimilar alleles (Prop. 2), vanishing as the panel around $a$ becomes dense and groove-tight.

### 4.3. Clustering and promiscuity

Thresholding $K_j$ (or the un-factored $K$) yields single-linkage allele *clusters* of functionally similar alleles (`Pseudoseq.cluster`). A peptide presented across a cluster is *promiscuous*; its promiscuity is the spread of its positive alleles over the clustering, and the cluster is the natural unit over which to pool nulls when per-allele data are thin (especially class II). This is the principled cross-allele similarity that the substrate omits.

## 5. Proteome-scale matching

**Presentation scan..** To find all presented peptides in a protein we slide every binding-length window (class I $8{-}11$; class II $\geq 13$ with the register-anchored core) and evaluate (3) per window and allele. With $W$ windows and $|\mathcal{A}|$ panel alleles the test is multiple: under the null the expected number of windows called for allele $a$ is $\approx W\alpha$, so we control the family either by Bonferroni / max-statistic thresholds (FWER) or by Benjamini–Hochberg on the $p_{\text{enrich}}$ across the $W \times |\mathcal{A}|$ grid (FDR) [5]. `scan_protein` reports per-window binders; the grid-level correction is a calibration step.

**Same-MHC versus TCR-facing search..** At proteome scale two notions of "similar peptide" are distinguished by which mask of (1) is indexed: *same-MHC* similarity indexes the presentation signature (peptides likely presented by the same allele), *TCR-facing* similarity indexes the anchor-masked central motif (similar T-cell recognition, the basis of cross-reactivity). Both run on the C++ `KmerIndex` seed-and-gather.

**Molecular mimicry..** A self or foreign peptide $p$ is a candidate cross-reactive mimic of a neoantigen $\nu$ iff (i) it is presented by a compatible allele, (ii) the TCR-facing distance $d_{\text{TCR}}(p,\nu) = \sum_i w_i \,\text{pen}(p_i,\nu_i) \leq \theta$ (anchors zeroed), and (iii) the hit is significant against the per-allele presented background, $\widehat{E} \leq \alpha$. Condition (iii) deflates motifs common in the allele's peptidome and elevates surprisingly shared ones. This connects to the neoantigen-quality program of Łuksza et al.: the quality $Q = R \times D$ [11, 10] multiplies a recognition / non-self term $R$ (the cross-reactivity distance to known antigens) by a self-discrimination term $D$ (differential MHC binding), and to the close-to-self "shell" picture of [15]; `mhcmatch` supplies a calibrated E-value for the $R$ component.

**Near-exact source identification..** Given a neoantigen we identify the self peptide it derives from—its parent protein and the mutated position—by full-sequence (unmasked) $\leq m$-mismatch search over all length-$L$ windows of the reference proteome. Under an i.i.d. background of composition $\{q_c\}$ the expected number of chance windows within $m$ substitutions of a query in a proteome of

$R$ residues is

$$\mathbb{E}[\#\text{chance}] \approx R \sum_{i=0}^{m} \binom{L}{i} \bar{q}^{\,L-i}(1-\bar{q})^{i}, \qquad \bar{q} = \Sigma_c\, q_c^2, \tag{8}$$

the Karlin–Altschul/birthday count specialized to the Hamming ball (the empirical-background variant reuses `seqtree` §KA). For $m \in \{0,1\}$ and $L \geq 8$, (8) is $\ll 1$, so a hit is essentially certainly the true source rather than coincidence—`Proteome.find_source` returns it with the differing residue flagged.

## 6. Motif logos and length distributions

For an allele's presented set (class I peptides of a fixed length; class II register-anchored 9-mer cores) the per-position information content is

$$I_i = \log_2 20 - H_i - e(n_i), \qquad H_i = -\sum_c f_{i,c} \log_2 f_{i,c}, \tag{9}$$

with $f_{i,c}$ the residue frequency at position $i$ and $e(n_i)$ the small-sample entropy correction for $n_i$ counts (negligible for the deep `pmhc_data` alleles). The logo draws letter $c$ at height $f_{i,c} I_i$ bits; columns at the anchors stand tall (e.g. A02:01 shows P2=L, PΩ ∈ {V,L}), the TCR-facing columns are flat. The length distribution is reported as a companion histogram. Columns may be confidence-weighted by $n$ (or by the shortlist $\geq$ 2-publication support). `logo.motif` returns the numeric logo; `logo.render` draws it.

## 7. Planned predictors and their composition

Presentation is necessary but not sufficient for immunogenicity. Each planned predictor scores a distinct biophysical or population term and *composes* with the presentation evidence on the log scale; for peptide $\sigma$ and allele $a$ the combined log-odds is

$$\log \frac{\mathbb{P}(\text{immunogenic})}{\mathbb{P}(\text{not})} = \beta_0 + \beta_{\text{pres}}\,(-\log p_{\text{enrich}}) + \beta_{\text{aff}} \log \tfrac{1}{\text{IC}_{50}} + \beta_{\text{stab}}\, t_{1/2} + \beta_{\text{clv}}\, c_{\text{Cterm}} + \beta_{\text{expr}} \log e_{\text{gene}} + \beta_{\text{imm}}\, \rho_{\text{TCR}}, \tag{10}$$

with coefficients fit on labelled outcome data (the user will supply tuning/benchmark sets). The terms: *affinity* $\text{IC}_{50}$ and *stability* $t_{1/2}$ are the quantitative complements of the binary presentation E-value, regressable on the same anchor features plus the pseudosequence; *cleavage* $c_{\text{Cterm}}$ is a position-specific C-terminal generation probability (with optional N-terminal trimming); *expression* $e_{\text{gene}}$ and variant frequency enter as Bayesian priors multiplying the presentation odds; *immunogenicity* $\rho_{\text{TCR}}$ combines physicochemical TCR-facing features with a TCR precursor-frequency / generation-probability estimate [12, 7] in the $Q = R \times D$ spirit of [11]. The precursor-frequency estimator may live in a separate package and be consumed here. Each term is a roadmap Phase-2 milestone specified by this subsection.

## 8. Benchmark and evaluation methodology

We evaluate as a per-(peptide, allele) binary task. *Splits* are by allele *and* by peptide: held-out alleles test cross-allele generalization (and the diffusion model of § 4), held-out peptides within an allele test within-allele recall; clonotype/ peptide duplicates are collapsed before splitting to prevent leakage. *Metrics* are ROC-AUC, PR-AUC against the prevalence baseline, top-1 accuracy, and a real-vs-random noise-rejection AUROC (the non-binder filter). *Comparison*: against NetMHCpan-4.1 and NetMHCIIpan-4.0 [14], MixMHCpred [4], and MixMHC2pred [13] on matched alleles and shared eluted-ligand / assay test sets, with rank-threshold calibration and an explicit *rare-allele subgroup* analysis (the equity axis of [8]) where the pseudosequence diffusion is expected to help most. The benchmark datasets and the paper live in a dedicated repository; this section fixes the protocol so it stays consistent with the predictors defined here.

## 9. Assumptions and limitations

(i) The analytic pooled null assumes approximate position independence in the anchor block; real co-occupancy adds a two-point ($b_2$) correction (the negative-binomial tail of the `seqtree` appendix) that keeps the mean robust while inflating the tail. (ii) Kernel validity rests on a faithful pseudosequence; ambiguous (`X`) positions are skipped and locus-aware normalization of allele names is required (class II especially). (iii) The pooled E-value is a smoothed estimator (Prop. 2)—its bias must be reported alongside the gain for rare alleles. (iv) The class-II register trick is a one-pass proxy for full deconvolution [3, 2]. (v) "Compatible allele" in mimicry is a user input; cross-reactivity significance is conditional on a faithful per-allele presented control. These are the open items tracked in `ROADMAP.md`.

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